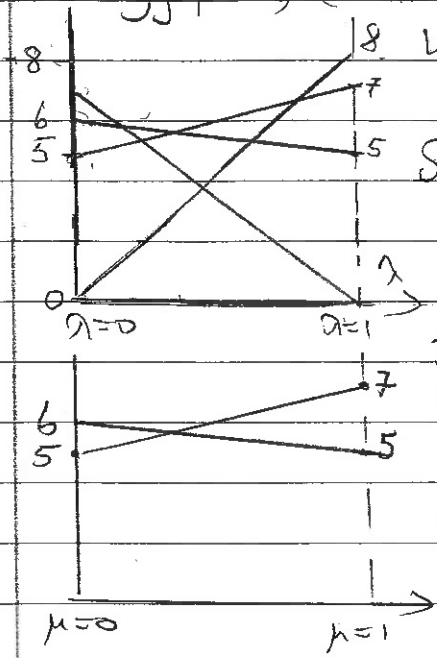
Model Answers for 2019 Game Theory Exam

1 (a) The pay-off matrix is $\begin{bmatrix} 6 & 5 & 7 & 0 \\ 5 & 7 & 0 & 8 \end{bmatrix}$. To calculate a safety strategy for A, $(1-\lambda)\times$row(1) $+\lambda\times$ row(2), we look at diagram:



Worst case pay-off (for A) is minimized for $6\times(1-\lambda)+5\lambda = 5\times(1-\lambda)+7\lambda \Rightarrow \lambda = \frac{1}{3}$.

So A's mixed strategy is

$\frac{2}{3}\times$ row(1) $+\frac{1}{3}\times$ row(2) —(1)

Because A chooses strategy (1), it will not be favorable for B to choose a mixed strategy involving either column (3) or (4), or both. So we may delete columns (3) and (4). Considering the reduced pay-off matrix $\begin{bmatrix} 6 & 5 \\ 5 & 7 \end{bmatrix}$, the worst case pay-off for B is maximized by the strategy $((1-\mu), \mu, 0, 0)$ when $6(1-\mu)+5\mu = 5(1-\mu)+7\mu \Rightarrow \mu = \frac{1}{3}$

Summary: a saddlepoint, in mixed strategies is $\left(\left(\frac{2}{3}, \frac{1}{3}\right), \left(\frac{2}{3}, \frac{1}{3}, 0, 0\right)\right)$ with pay-off $\frac{17}{3}$ —(2)

(b) Let $(\bar{y}, \bar{z})$ be a saddlepoint. Then $\bar{y}^T S z \le \bar{y}^T S \bar{z} = V \le y^T S \bar{z}$ for all y, z.

Let $\bar{p} = -V$ and $\bar{q} = +V$. Then, from (2),

$\bar{y}^T S z \le \bar{q}$, for all $z \in \Lambda_m \Rightarrow \bar{y}^T S \le \bar{q}\, 1_m^T$ —(3)

$y^T S \bar{z} \ge -\bar{p}$ for all $y \in \Lambda_n \Rightarrow S\bar{z} \ge -\bar{p}\, 1_n$ —(4)

(3) and (4) imply $(\bar{y}, \bar{z}, \bar{p}, \bar{q})$ is feasible for (P)

Clearly the pay-off is $\bar{p}+\bar{q} = -V+V = 0$ —(5)

Take any other feasible point (y, z, p, q). (6)

$Sz \ge -p\, 1_n \Rightarrow y^T S z \ge -p\, y^T 1_n = -p \Rightarrow y^T S z \ge -p$

$y^T S \le q\, 1_m^T \Rightarrow y^T S z \le q\, 1_m^T z = q \Rightarrow y^T S z \le q$

(6) implies $p+q \ge 0$

Thus, the pay-off for $(y, z, p, q) \ge 0$.

Combining this relation with (5), we conclude $(\bar{y}, \bar{z}, \bar{p}, \bar{q})$ is a minimizer for (P).

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2. The possible strategies for B are

$B_1 = \begin{cases} L \text{ if } A=L \text{ and } N=\text{'no fault'} \\ L \text{ if } A=R \text{ and } N=\text{'no fault'} \\ L \text{ if } N=\text{fault} \end{cases}$, $B_2 = \begin{cases} L \\ R \\ L \end{cases}$ if... , $B_3 = \begin{cases} R \\ L \\ L \end{cases}$, $B_4 = \begin{cases} R \\ R \\ L \end{cases}$.

$B_5 = \begin{cases} L \\ L \\ R \end{cases}$, $B_6 = \begin{cases} L \\ R \\ R \end{cases}$, $B_7 = \begin{cases} R \\ L \\ R \end{cases}$, $B_8 = \begin{cases} R \\ R \\ R \end{cases}$

If N = 'no fault' ('fault'), then the strategies are

A \ B	B1	B2	B3	B4	B5	B6	B7	B8
L	LL (LL)	LL (LL)	LR (LL)	LR (LL)	LL (LR)	LL (LR)	LR (LR)	LR (LR)
R	RL (RL)	RR (RL)	RL (RL)	RR (RL)	RL (RR)	RR (RR)	RL (RR)	RR (RR)

We can then obtain the pay-offs from the strategy 'tree'.

A \ B	B1	B2	B3	B4	B5	B6	B7	B8	
L	2	2	1	1	2	2	1	1	'no fault'
R	1	0	1	0	1	0	1	0	

A \ B	B1	B2	B3	B4	B5	B6	B7	B8	
L	2	2	2	2	1	1	1	1	'fault'
R	1	1	1	1	0	0	0	0	

0.5 x 'no fault' + 0.5 x 'fault' gives pay-off:

pay-off	B1	B2	B3	B4	B5	B6	B7	B8
$\frac{1}{2}$ x L	4	4	3	3	3	3	2	2
$\frac{1}{2}$ R	(2)	1	(2)	1	1	0	1	0

(both admissible)

This has 2 Nash Equilibria: (R, B_1) and (R, B_3)

Pay-offs are $\frac{1}{2}$ in both cases

Notice that, if A chooses R: (fault)

'B_1' means B chooses L (if, or not, there is a fault)

and 'B_3' " " also choose L (" " ")

So B_1 and B_3 give the same action (L) for B.

Thus there is one Nash equilibrium: (R, L)

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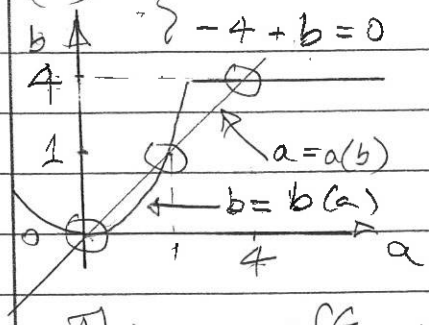
3. $a \to L^A(a,b)$ is convex, so the minimizing a is given by $\frac{\partial}{\partial a} L^A(a,b) = 0$

$b \to L^B(a,b)$ " " " " b " " $\frac{\partial}{\partial b} L^B(a,b) = 0$

Hence the response curves $a(b)$ and $b(a)$ for A and B are

(A): $-b + a = 0 \Rightarrow \underline{a(b) = b}$ and

(B): $\begin{cases} -a^2 + b = 0 & \text{if } a \le 2 \\ -4 + b = 0 & \text{if } 2 < a \end{cases} \Rightarrow b(a) = \begin{cases} a^2 & \text{if } a \le 2 \\ 4 & \text{if } a > 2 \end{cases}$



We see that there are 3 Nash equilibria

$(\bar{a}_1, \bar{b}_1) = (0,0)$, $(\bar{a}_2, \bar{b}_2) = (1,1)$

and $(\bar{a}_3, \bar{b}_3) = (4,4)$. (*)

The pay-offs are $\underline{(0,0)}$, $\underline{(-\tfrac{1}{2}, -\tfrac{1}{2})}$, $\underline{(-8,-8)}$.

For the Stackelberg game, B plays $b(a)$.

So A minimizes

$$\psi(a) = L^A(a, b(a)) = \begin{cases} -a^3 + \frac{1}{2}a^2 & \text{if } a \le 2 \\ -4a + \frac{1}{2}a^2 & \text{if } a > 2 \end{cases}$$

Notice that $\frac{d}{da}\psi(a) = 0$ when $a = 0, \frac{1}{3}$ and 4

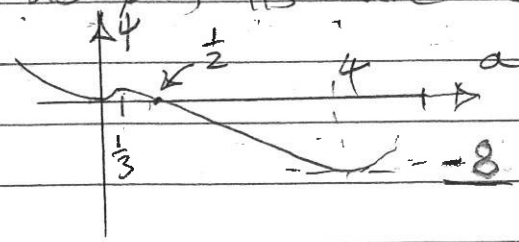
Then $\psi(0) = 0$, $\psi(\frac{1}{3}) = \frac{1}{9}(\frac{1}{2} - \frac{1}{3}) > 0$, $\psi(4) = -8$

It follows that $\psi(a)$ has a unique minimum at $\bar{a} = 4$.

Thus $\underline{\bar{a} = 4}$ is the unique Stackelberg optimizer for A.

The corresponding optimizing strategy for B is $\underline{\bar{b} = 4}$

The pay-offs are $L^A(\bar{a}, \bar{b}) = \underline{-8}$ and $L^B(\bar{a}, \bar{b}) = \underline{-8}$



Graph of $\psi(a)$ (not to scale)

Observe that the unique Stackelberg optimizing pair $(\bar{a}, \bar{b})$ coincides with the most "efficient" Nash equilibrium.

In (*), $(-8,-8)$ is the best because it gives lowest pay-off for both players

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4. Consider X_1's pay off. There are 3 relevant scenarios
$S_1 = \{X_1 \text{ sails}\}$, $S_2 = \{X_1 \text{ not sail but either } X_2 \text{ or } X_3 \text{ sails}\}$
Assume and $S_3 = \{\text{nobody sails}\}$
S_1 chooses $((1-\lambda_1), \lambda_1)'$ for not sail / sail and
S_j " $((1-\lambda_j), \lambda_j)$ " " , for $j = 2, 3$.
Then $P(S_1) = \lambda_1$, $P(S_2) = (1-\lambda_1)(1-(1-\bar{\lambda}_2)(1-\bar{\lambda}_3))$
and $P(S_3) = \lambda_1 \bar{\lambda}_2 \bar{\lambda}_3$.
Since S_1, S_2, S_3 have costs $(1-c_1), 1, 0$ respectively.
So pay-off for $X_1 = (1-c_1)\lambda_1 + (1-\lambda_1)(\bar{\lambda}_2 + \bar{\lambda}_3 - \bar{\lambda}_2\bar{\lambda}_3) - \lambda_1\bar{\lambda}_2\bar{\lambda}_3$
Likewise,
pay-off for $X_2 = (1-c_2)\lambda_2 + (1-\lambda_2)(\bar{\lambda}_1 + \bar{\lambda}_3 - \bar{\lambda}_1\bar{\lambda}_3) - \lambda_2\bar{\lambda}_1\bar{\lambda}_3$
pay-off for $X_3 = (1-c_3)\lambda_3 + (1-\lambda_3)(\bar{\lambda}_1 + \lambda_2 - \bar{\lambda}_1\bar{\lambda}_2) + \lambda_3\bar{\lambda}_1\bar{\lambda}_2$
For these pay-offs to be minimized at a strict mixed strategy,
$2\bar{\lambda}_2\bar{\lambda}_3 =$
Hence $\begin{cases} 2\bar{\lambda}_2\bar{\lambda}_3 = \bar{\lambda}_2 + \bar{\lambda}_3 - (1-c_1) \\ 2\bar{\lambda}_1\bar{\lambda}_3 = \bar{\lambda}_1 + \bar{\lambda}_3 - (1-c_2) \\ 2\bar{\lambda}_1\bar{\lambda}_2 = \bar{\lambda}_1 + \bar{\lambda}_2 - (1-c_3) \end{cases}$ ← 3 equations. Likewise and
Now assume $c_1 = c_2 = c_3 = c$
By symmetry, $\bar{\lambda}_1 = \bar{\lambda}_2 = \bar{\lambda}_3 := \bar{\lambda}$. We have
$$\bar{\lambda}^2 - \bar{\lambda} + (1-c)/2 = 0 \quad \text{—— (1)}$$
This has 'real' solutions
$$\bar{\lambda} = \frac{1}{2}\left(1 \pm \sqrt{2c-1}\right)$$
since $\frac{1}{2} < c < 1$. Thus the Nash equilibria are $((1-\bar{\lambda}), \bar{\lambda})$ (for each player, where)
$$\bar{\lambda} = \frac{1}{2}\left(1 + \sqrt{2c-1}\right) \text{ or } \bar{\lambda} = \frac{1}{2}\left(1 - \sqrt{2c-1}\right).$$
The probability that no-one will sail is $(1-\bar{\lambda})^3$ (two possible values)
If $0 < c < \frac{1}{2}$ there are no strictly mixed Nash equilibria, but "everyone sails" is a Nash equilibrium in pure strategies.

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(A) The DP approach is to consider the Bellman-Isaac equation

$$\begin{cases} V_t(x) = \min_u \max_v \{ V_{t+1}(f(x,u,v)) + L(x,u,v) \} \\ V_N(x) = g(x) \end{cases}$$

(*)

This is solved backwards in time, using the boundary condition. It is assumed at each iteration the 'value' $\min_u \max_v \{(*)\}$ is well-defined. Knowing $V_t(x)$, $t = N, N-1, \ldots, 0$, we get the saddle point $(\bar{u}_t(x), \bar{v}_t(x))$ from $(\bar{u}_t(x), \bar{v}_t(x))$ is a saddle point for (*).

5(B) Following hint, assume $V_t(x) = p_t x^6$ for $t = 0, \ldots, N$. Then $p_N x^6 \equiv x^6$, so $p_N = 1$.

$$p_t x^6 = \min_u \max_v \{ p_{t+1}(x^3 + u + v)^2 + u^2 - \gamma v^2 \}$$

To get saddlepoint (u, v) set $\partial/\partial u \{\ldots\} = \partial/\partial v \{\ldots\} = 0$. This gives

$2p_{t+1}(x^3 + u + v) + 2u = 0$ and $2p_{t+1}(x^3 + u + v) - 2\gamma v = 0$

Hence $u = -\gamma v$ and $p_{t+1} x^3 + (p_{t+1}(1-\gamma^{-1}) + 1)u = 0$. Hence

$$u_t(x) = -\frac{p_{t+1} \cdot x^3}{1 + p_{t+1}(1-\gamma^{-1})}, \quad v_t(x) = \frac{\gamma^{-1} p_{t+1} \cdot x^3}{1 + p_{t+1}(1-\gamma^{-1})} \quad \text{--- (1)}$$

Substituting into $\{\ldots\}$ gives

$$p_t x^6 = p_{t+1} x^6 \left(\frac{1}{1 + p_{t+1}(1-\gamma^{-1})}\right)^2 + \frac{(1-\gamma^{-1}) p_{t+1}^2 x^6}{(1 + p_{t+1}(1-\gamma^{-1}))^2}$$

$$= p_{t+1} x^6 \times \frac{1 + (1-\gamma^{-1}) p_{t+1}}{(1 + (1-\gamma^{-1}) p_{t+1})^2} = \frac{p_{t+1} x^6}{1 + (1-\gamma^{-1}) p_{t+1}}$$

Since this must be true for all x,

$$p_t = \frac{p_{t+1}}{1 + (1-\gamma^{-1}) p_{t+1}} \quad \text{--- (2)}$$

Summarizing: the p_t's are obtained by solving (2) backwards, with starting condition $p_N = 1$. Then the saddle point controls

$$(u_t(x), v_t(x))$$

are given by (1).